

Midland Energy Resources, Inc. – Cost of Capital Analysis

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Our analysis began with a fundamental observation: Midland Energy Resources cannot be evaluated as a single, homogenous enterprise. The firm's three core divisions: Exploration & Production (E&P), Refining & Marketing (R&M), and Petrochemicals operate in distinct markets, face different sources of risk, and utilize capital in different ways. Applying one uniform cost of capital across such diverse businesses distorts project evaluations, misprices risk, and leads to inconsistent capital allocation across divisions.

Using one rate across different business models causes high-risk projects to appear more attractive and low-risk ones to appear unprofitable. A single WACC ignores the economic reality of our operations and undermines shareholder value.

By calculating the cost of capital for each division and weighing them by the relative assets of each business, we can produce a more accurate financial benchmark for project valuation. This approach captures the true cost of financing, aligns performance expectations with divisional risk, and supports investment strategies that promote long-term value creation. Midland's cost of capital estimates are not merely theoretical figures; they function as the company's financial control tools, including serving as hurdle rates for capital budgeting, capital charge rates within the Economic Value Added (EVA) framework, and discount rates for valuation. Our analysis applied consistent corporate assumptions: a 30-year Treasury yield of 4.98% as the long-term risk-free rate for E&P and R&M, and a 10-year yield of 4.66% for Petrochemicals to reflect its shorter investment horizon, along with a 5.00% market risk premium and a 39.73% tax rate, while adjusting for each division's leverage, credit spread, and operating risk. These assumptions reflect both Midland's financial position and prevailing market conditions in early 2007. The 30-year U.S. Treasury yield of 4.98% provides a stable long-term benchmark consistent with Midland's investment horizon for major capital projects. The 5.00% equity market risk premium aligns with the company's internal policy and historical averages observed in mature markets, balancing analyst estimates between 4.00% and 6.00%. Together, these inputs provide a realistic and internally consistent foundation for estimating Midland's divisional and corporate WACC.

- **Exploration & Production (E&P)** represents Midland's most capital-intensive and volatile segment. Using comparable companies within the upstream energy sector, we unlevered the observed equity betas of four peers to remove the impact of their differing leverage levels, yielding an average unlevered beta of 0.93. Based on the division's 1.6 % credit spread over Treasuries and a 5 % market risk premium, we computed a debt beta ($\beta_{\text{Debt}} = 0.32$) to reflect the moderate market sensitivity of Midland's investment-grade debt. This was then re-levered at 46.00% target debt-to-value ratio, resulting in an equity beta of 1.25. Applying to the CAPM with a 4.98% risk-free rate and a 5.00% market risk premium produced a cost of equity of 11.21%, which, when combined with the after-tax cost of debt of 3.97%, resulted in a divisional WACC of 7.88%. (Exhibit 2)
- **Refining & Marketing (R&M)** is exposed to margin cycles and commodity spreads. The division maintains a moderate leverage level of 31.00% debt to total value and a BBB credit rating. Using a 4.98% long-term Treasury yield as the risk-free rate and a 5.00% equity market risk premium, a beta of debt of 0.36, division's cost of equity was estimated at 11.16%. The cost of debt, derived from the 1.80% spread over Treasuries and adjusted taxes at 39.73%,

yields an after-tax cost of 4.09%. Combining these inputs results in a divisional WACC of 8.97%, reflecting the higher cyclicalality and business risk inherent to the refining segment. (Exhibit 3)

- **Petrochemical** is Midland's smallest but steady-earning division, supported by diversified chemical products and joint ventures with moderate risk exposure. Owing to its lower volatility and shorter project cycles, a 10-year Treasury yield of 4.66% was used as the risk-free rate, better reflecting its investment horizon than the 30-year benchmark applied elsewhere. With a divisional beta of 1.30 computed based on the weighted average of each division's leveraged beta in Midland's leveraged beta of 1.25, a 5.00% market risk premium, and a 40.00% debt-to-value ratio, the cost of equity was estimated at 11.15% and the after-tax cost of debt at 3.62%, yielding a divisional WACC of 8.14% that aligns with its stable cash flows and conservative capital structure. (Exhibit 4)

The divisional WACC serve as hurdle rates for investment approval, capital charge rates in the Economic Value Added Tax (EVA) framework, and discount rates for valuing the firm and its equity. Because these figures guide real capital deployment and managerial accountability, they more accurately reflect each of the three division's operating risk, leverage, and credit grade.

It is not financially appropriate for Midland to stick to a single corporate WACC for overall valuation. Each division operates under a fundamentally different risk return profile. Exploration & Production (E&P)'s high exposure to commodity prices and capital intensity justifies a higher hurdle rate. Refining & Marketing (R&M)'s cyclicalality and competitive volatility elevate its cost of capital while Petrochemical's steady income and strong credit quality results in a lower rate compared to the other two divisions.

If we were to apply a flat corporate WACC, there would be systematic distortion into project evaluations. High-risk projects in Refining & Marketing (R&M) will be overvalued, encouraging overinvestment in volatile operations. Low-risk projects in Petrochemicals will be undervalued, leading to underinvestment in stable, profitable opportunities.

By adopting divisional WACC rates, Midland can eliminate these flaws and inefficiencies. The divisional WACC framework ensures that each business is evaluated against a return standard that accurately reflects its risk profile. This gives management the clarity to prioritize capital toward the highest value creating projects. We recommend the application of the divisional WACC structure for internal planning, investment project appraisals, and performance measurements. The proposed WACC rates: 7.88% for E&P, 8.97% for R&M, and 8.14% for Petrochemicals should be the divisional benchmark for project evaluation and capital budgeting. The consolidated 8.30% firmwide WACC rate can remain as the reference point for corporate-level financing, mergers, and valuation.

This approach safeguards against capital misallocation and strengthens the relationship between risk and return. This positions Midland to deliver sustained profitability and superior long-term shareholder value.

APPENDIX

Exhibit 1: Divisional WACC and Financial Profile Summary

Division	Target D/V	Credit Rating	Cost of Equity	After-Tax Cost of Debt	WACC
Exploration & Production	46.00%	A+	11.21%	3.97%	7.88%
Refining & Marketing	31.00%	BBB	11.16%	4.09 %	8.97%
Petrochemicals	40.00%	AA–	11.15%	3.62%	8.14%
Consolidated (Asset-Weighted)	42.20%	A+	—	—	8.30%

The corporate tax rate of 39.73% was derived from Midland's effective tax rates over 2004–2006, ensuring consistency with reported financial performance.

Exhibit 2: Exploration & Production – Cost of Capital Computation

Peers	Equity Beta	Debt/Equity	Unlevered Beta
Jackson Energy, Inc.	0.89	11.20%	0.83
Wide Plain Petroleum	1.21	85.40%	0.80
Corsicana Energy Corp.	1.11	15.20%	1.02
Worthington Petroleum	1.39	47.50%	1.08
Average Beta	-	-	0.93

Unlevered and relevered betas derived from comparable firms to estimate divisional cost of capital for E&P.

Debt / Equity (B/S)	0.85
Divisional Equity Beta	1.25
Cost of Equity	11.21%
Cost of Debt (After-Tax)	3.97%
WACC	7.88%

Exhibit 3: Refining & Marketing – Cost of Capital Computation

Peers	Equity Beta	Debt/Equity	Unlevered Beta
Bexar Energy, Inc.	1.70	10.30%	1.60
Kirk Corp.	0.94	19.40%	0.84
White Point Energy	1.78	20.90%	1.58
Petrarch Fuel Services	0.24	-12.00%	0.26
Arkana Petroleum Corp.	1.25	32.30%	1.05
Beaumont Energy, Inc.	1.04	20.60%	0.93
Dameron Fuel Services	1.42	50.30%	1.09
Average	-	-	1.05

Unlevered and relevered betas derived from comparable firms to estimate the divisional cost of capital for R&M.

Debt / Equity (B/S)	44.93%
Divisional Equity Beta	1.24
Cost of Equity	11.16%
Beta of Debt	0.36
Cost of Debt (After-Tax)	4.09%
WACC	8.97%

Exhibit 4: Petrochemicals – Cost of Capital Computation

	E&P	R&M	Petrochemicals	Midland
Total Assets (2006)	\$140,100	\$93,829	\$28,450	262,378
Weight of Division	53.40%	35.76%	10.84%	100%
Levered Beta (Weighted across divisions)	1.25	1.24	1.30	1.25

Weighted-average beta approach used to impute Petrochemicals' cost of capital in absence of direct peer data.

Debt / Equity (B/S)	66.67%
Divisional Equity Beta	1.30
Cost of Equity	11.15%
Cost of Debt (After-Tax)	3.62%
WACC	8.14%

A 10-year U.S. Treasury yield (4.66 %) was used as the risk-free rate to reflect Petrochemicals' shorter project duration and lower risk profile relative to E&P and R&M.

Exhibit 5: Risk-Adjusted WACC: Internal Capital Allocation Impact

Division	Hurdle rate (true WACC)	Implied risk level	Capital Allocation Impact
E&P	7.88%	Moderate	Minor misallocation -> maintain capital discipline for the core business
R&M	8.97%	Volatile	Over-investment -> value destruction (high risk-high return)
Petrochemicals	8.14%	Stable	Under-investment -> value-preserving assets

E&P is moderate with minor misallocation, R&M is volatile with risk of over-investment, Petrochemicals is stable but may under-invest

Exhibit 6: Treasury yield curve for risk-free rate

Maturity	1-Year	10-Year	30-Year
Rate	4.54%	4.66%	4.98%

Exhibit 7: Credit rating and debt spreads by segment

Business Segment	Credit Rating	Target Debt/Value	Spread over Treasury
Consolidated	A+	42.20%	1.62%
E&P	A+	46.00%	1.60%
R&M	BBB	31.00%	1.80%
Petrochemicals	AA-	40.00%	1.35%

Higher-rated segments have lower spreads, lower-rated segments have higher spreads

Exhibit 8: Re-arranged balance sheet of 2006

	(\$ millions)
Notes Receivable	19,68
Inventory	7,29
Prepaid Expenses	2,23
Operating Current Assets Total	29,19
Accounts Payable & Accrued Liabilities	(26,58)
Deferred Taxes (Current)	(5,46)
Net Working Capital (NWC)	(2,85)
Net Property, Plant & Equipment	167,35
Investments & Advances	9,29
Other Assets	9,47
Post Retirement Benefit Obligations	(4,84)
Accrued Liabilities (Long-Term)	(14,18)
Deferred Taxes (Long-Term)	(2,73)
Other Long-Term Liabilities	(179,63)
Net Operating Assets	176,79)

	\$ millions
Current Portion of Long-Term Debt	20,78
Long-Term Debt	81,08
Less: Cash and Equivalents	(19,21)
Less: Restricted Cash	(3,13)
Net Debt	79,51
Book Value	97,29
Net Debt + Equity	176,79

Re-arranged 2006 balance sheet shows net operating assets of \$176.79M, funded by \$79.51M net debt and \$97.29M equity

Exhibit 8: Capital Structure Summary

	\$ billions	% of Total Capital
Market Value of Equity (E)	134.09	62.80
Market Value of Debt (D)	79.54	37.20%
Total Firm Value (V)	213.63	100.00%
D/E Ratio	0.59	59.30%
D/V Ratio	0.37	37.20%
E/V Ratio	0.63	62.80%
Net Debt (book-based)	79.50	—
Shareholders' Equity (book)	97.30	—
Net Capital	176.80	—

Midland relies mainly on equity financing with moderate leverage, supporting financial stability